Exploring Generative Artificial Intelligence in Upper Secondary Art and Design Education: Developing Digital Competence and Critical Reflection

Thelma Andrine Helland Gustavsen, Sofie Fliflet Sollie, Ingri Strand and Tore Andre Ringvold,
Oslo Metropolitan University,
Thelmaahg@icloud.com and Sofiesollie@gmail.com.

Abstract

This project investigates how generative artificial intelligence (AI) can be integrated into upper secondary art and design education, to enhance students' digital competence, critical reflection, and digital judgment. The study is set within a contemporary context where generative AI is increasingly shaping education, work, and society. The Norwegian curriculum emphasizes digital skills as a core competency, highlighting this project's relevance for today's educational landscape. The aim of this project is to explore how teaching and learning with generative AI, can foster creative, ethical, and reflective practices in design education. By embedding AI tools into the classroom, the project seeks to investigate how this technology can serve as a resource to spark students' creativity, while also addressing ethical challenges and encouraging critical evaluation of AI-generated outputs. Methodologically, the project applies educational design research (EDR), combining theoretical development with practical interventions. We designed an assignment where students use generative AI to create an advertising poster, focusing on visual devices, target audience, content and meaning. We used the qualitative methods unstructured classroom observation, semi-structured interviews and analysis of student work. 22 students, in two groups, and one teacher take part in two iterative design cycles. Thematic analysis is used to identify patterns in interaction, reflection, and learning among the participants. This project contributes to a further understanding of how generative AI can be adapted to educational contexts to strengthen students' digital skills and critical thinking. It also sheds light on how teachers can facilitate meaningful learning experiences in response to a rapidly evolving digital world. By sharing insights from this research, we aim to contribute to broader discussions on how AI can be integrated in education in ways that promote creativity, ethical and critical reflection.

Key Words: AI Literacy, Generative AI, Digital Competence

1 INTRODUCTION

Generative artificial intelligence (AI) is an emerging technology in today's education, shaping how students learn, create, and make decisions. This places new demands on their abilities in critical evaluation and influences which competencies they may need in their future education, work, and participation in society. In educational contexts, generative AI tools that produce visual content challenge established understandings and raise new questions about creativity, authorship, and critical evaluation, highlighting the need for reflective and responsible approaches to AI supported tools.

As generative AI increasingly enters educational settings—intentionally or otherwise—there is a growing need to examine how students develop an understanding of how these technologies function, how they may be used as creative resources, and which ethical and critical considerations are involved. This need extends beyond individual competencies to include how students negotiate meaning, make decisions within iterative processes, and engage in collaborative reflection when working with AI-supported tools.

Drawing on research conducted for our master's thesis within a Norwegian upper secondary art, design, and architecture programme, this paper examines how students develop AI and digital literacy through the use of generative AI and collaborative group work, where AI functions as a shared reference point for dialogue, evaluation, and learning.

2 LITERATURE REVIEW

Building on Vartiainen et al.'s research on co-creation with generative AI and peer interactions (Vartiainen et al., 2023), this study extends their work through a practical educational design that emphasizes dialogic teaching, iterative prompting and ethical reflection to evaluate its impact on students' AI and digital literacy. In line with research on AI in education, Heaton et al. (2024) argue for pedagogical approaches that position students as active and reflective participants in AI-supported creative processes, rather than passive users of generative technologies. Together, this perspective informs the educational design in the present study, with a focus on students' development of AI knowledge and critical reflection on data use, consent, privacy, and responsibility.

3 THEORETICAL FRAMEWORK

3.1 Pragmatism

The project draws on John Dewey's pragmatist concept of knowledge, which conceptualises learning as developing through practice, controlled experience, action, and reflection (Dewey, 1916). Dewey regarded experience as the basis for all knowledge, and this experience comes from a constant transaction with the environment (Dewey, 1929, 1938). From this perspective, learning emerges through active engagement with concrete situations and reflection on the consequences of one´s actions. This perspective underpins the research project's iterative process, where

teaching is continuously adapted based on empirical experiences to support students' development of digital skills and critical thinking. The link between experience and knowledge is also reflected in the teaching design, where the students shift between working with generative AI and reflecting on what they have experienced.

Dewey worried that students would learn subject matter isolated from real life (Dewey, 1916) and stressed that education should provide experiences relevant outside of educational institutions, that can be connected to further experiences and arouse the learner's curiosity and desire to learn more (Dewey, 1938). The design is therefore grounded in current and societally relevant issues that students encounter in contemporary society and is structured to facilitate collaboration, exploration, evaluation, and problem-solving. This approach aligns with Dewey's understanding of learning as a problem-solving process that emerges in response to challenges and develops through active participation, dialogue, and collaboration (Bailey et al., 2010).

3.2 AI literacy and digital literacy

This study is grounded in AI-literacy, understood as the ability to collaborate and communicate with AI while engaging with it critically and ethically (Long & Magerko, 2020). The design for the project was also anchored in the Norwegian Directorate of Education and Trainings framework for students' digital literacy, which emphasizes privacy, source evaluation, ethics, and copyright when using digital media (Norwegian Directorate for Education and Training, 2017). By embedding these frameworks into classroom activities, the project investigates how generative AI can function as a resource in students' creative processes, while also addressing ethical challenges and encouraging critical evaluation and AI-generated outputs.

4 METHODOLOGY

4.1 Participants

The participants were students from the Norwegian upper secondary art, design, and architecture program, which grants general university entrance qualifications. The program focuses on creative work and developing skills in visual art, design, and architecture. One teacher carried out the educational design by facilitating dialogue during presentation and guided students as they worked with the assignment. We did not focus on the teacher's implementation, as the focus was on the students.

A total of 22 students participated in the study. They were split into two groups, the first intervention had a group of 10 students, and the second intervention had 12 students.

4.2 Ethical Considerations

This research project has been approved by Sikt – Norwegian Agency for Shared Services in Education and Research (SIKT, n.d.). This ensures students and teachers privacy, anonymity, and data protection. The teacher and students have signed to declare informed consent to participate.

4.3 Educational design research (EDR)

This study's methodological approach is educational design research (EDR), which aims to address and develop solutions to challenges in educational practice, while contributing to theory development (McKenney & Reeves, 2018). We conducted the educational design research project through two iterative cycles of design, implementation, evaluation and re-design before trying it again, collecting data during the implementation phase.

4.3.1 The Components of the educational design

The design consisted of:

- Presentation material: A PowerPoint introduced generative AI, technical use, privacy considerations, and contemporary applications of AI in advertising. It addressed strategies for identifying AI-generated content and ethical issues related to responsibility, transparency, misinformation, labelling, and applicable legal regulations, supported by integrated questions to promote critical discussion.
- Assignment and worksheet: students collaborated in pairs to design an advertising poster, combining analogue sketching with AI-based image generation (first text-to-image and then image-to-image). Their design process, including ideas, sketches, prompts, and outcomes, was documented in a worksheet.

4.4 Data collection and analysis

Educational design research commonly employs qualitative methods for collecting and analysing data (McKenney & Reeves, 2014). We have chosen the qualitative methods observation and interview, presented in the following sections.

4.4.1 Observation

Observation was chosen as a method to examine how the design worked, what could be improved, and if it fostered students' digital literacy, and AI-literacy, through observing interactions and practical work. We observed one instructional week per intervention and had an unstructured approach. The first intervention totalled 8 school hours (45 min. each) and the second 9 school hours, both spread across the week. Our observational roles were non-participant (Bryman, 2016). Right after the lessons, we made field notes describing what we had observed.

4.4.2 Interview and group discussion

Two types of interviews were conducted with the students to collect data:

- Semi-structured interviews (Brinkmann & Kvale, 2015): Focused on assessing the educational design, including what functioned well and what could be improved.

- Group discussion: Explored students' experiences, the technical skills they developed, their ethical reflections on AI, and their perceptions of artefacts, bias, and limitations in AI-generated outputs. Both part of the design and as a way to gather data.

Drawing on the group discussion, this paper primarily examines students' ethical reflections on AI, with particular attention to privacy, copyright, and misinformation, which are commonly understood as key components of AI literacy and digital literacy.

4.4.3 Thematic analysis

We used thematic analysis based on Braun and Clarke's (2006) framework as our qualitative approach, due to its focus on identifying recurring themes across the dataset. This framework was used to analyse meaning in interview transcriptions, fieldnotes, and students' work sheets. Themes were developed with the research question in mind. The analysis involved familiarizing with the data, generating initial codes, developing, reviewing and defining themes, and reporting the results in relation to theory.

The analysis was conducted inductively, and four themes developed:

- Digital literacy
- Student engagement
- Technical skills with AI
- Redesign

In this paper we will present and elaborate on the results from the themes *Digital literacy* and *Redesign*. Due to word limitations, the themes are not fully explored.

4.5 Disclosure on the use of Generative AI

The data analysis and text production for this paper is done entirely by the authors, but GenAI-tools have been used in translations from Norwegian to English and for copy editing.

5 RESULTS

Here we are presenting results from intervention 1 and 2, focusing on Digital literacy and what had to be redesigned between and after each intervention.

5.1 Results from first intervention: Digital literacy

5.1.1 Students' reflections on copyright

Copyright emerged as a key theme in intervention 1. Observations indicated that several students raised questions regarding copyright during the lessons and engaged in discussions concerning

the ethical frameworks governing the use of AI for image production, as well as whether such practices comply with established guidelines.

During the group discussion, concerns and considerations emerged regarding the origins of the images generated by AI and copyright infringement in the training material. One student (Student 3) expressed uncertainty about whether all images used to train AI systems have been used with consent from the original creators and questioned the ethical implications of such use. This statement may indicate that the student is reflecting on potential issues related to ethical use of generative AI.

The student groups approached copyright issues differently in their own work. One group attempted to generate a well-known brand logo in their advertisements. After initial challenges, they succeeded, which may indicate that the AI systems enforce copyright protection and trademarks only to a limited extend. The group commented on their submission, which may be interpreted as an awareness of copyright issues and as emerging understanding of the limitations and potential workarounds within the AI system.

In contrast, another group described a conscious effort to uphold copyright in their work. They refrained from using copyright-protected images from existing brands and instead drew inspiration from a magazine cover, while deliberately avoiding providing the AI with photographs taken by others. These reflections are also articulated in their submitted work.

5.1.2 Students' reflections on privacy

During the group discussion, students reflected on privacy issues in AI use and whether they had considered privacy when completing the task. An analysis of the submitted task sheets revealed no prompts that raised privacy concerns. Several students also reported deliberately avoiding the use of images of themselves or others, as well as personal data. This suggests that the teaching design adequately addresses privacy issues and that students demonstrated awareness of privacy considerations when using generative AI.

Although the students took privacy considerations into account, they also expressed concerns regarding the training material, and whether this may include images of individuals who have not given consent. One student group noted that they had generated a face that appeared familiar. Another student highlighted children's privacy as a particular area of concern, pointing out that publicly available images may potentially be misused as AI training data.

5.1.3 Students' reflections on misinformation in AI-generated content

In both the observation and the group discussion, students explained why and how the use of AI should be labelled, and how unlabelled AI images may affect consumers' trust and perceived value. Several students were surprised that clear regulations for AI-generated advertising have not yet been established, particularly given existing requirements for labelling digital retouching in advertising. They emphasised that distinctions between what is real and artificial can be

blurred, potentially leading to uncertainty and unrealistic expectations, especially in contexts such as cosmetic advertising.

5.2 Redesign after the first intervention

5.2.1 Including copyright

We initially removed slides on copyright from the first intervention per the teachers' advice, but students repeatedly raised questions on AI and ethics on copyright. We therefore reinstated a copyright component within the AI and ethics module in the PowerPoint.

5.2.2 Instructional design: Assignment execution and Structure

Many students generated before ideating or sketching and underinvested in sketch development, which hindered Assignment 2 and caused frustration. We emphasized that stronger sketches lead to better outputs and implemented a phased workflow:

(i) Idea and design development with sketching, documented in assignment sheet
(ii) Assignment 1: Prompting based on sketches (text-to-image) + documenting process.
(iii) Assignment 2: Using sketch image as input in the prompt (image-to-image) + documenting process.

5.3 Results from the second intervention: Digital literacy

5.3.1 Students' reflections on copyright

In contrast to the first intervention, several students wanted to design advertisements for well-known brands using trademarked brands and logos. During image generation, Microsoft Co-pilot sometimes refused to generate content due to copyright restrictions, while at other times it allowed protected logos to be generated. This inconsistency triggered several students to discuss what is acceptable and how one should relate to copyright if AI does not enforce it. In group work, one student said it felt wrong that one person had access and another didn't, the same group continued, reasoning that: 'AI doesn't care about copyright anyway', and therefore it was fine to continue. However, two of the other groups chose to reconsider their design choices and their prompting strategies during their design process, to maintain a certain originality. Asked what they could do to uphold copyright after experiencing the AI tool's inconsistency, the students expressed different strategies. One student stated that to use generative AI ethically, one must create entirely new designs and steer clear of existing names or logos and reported doing so. Another student favoured changing the name while keeping the look but later switched to an original design to avoid copyright risk.

5.3.2 Students' reflections on privacy

In the group discussion, the students were asked whether they had adhered to established privacy guidelines. They confirmed that they had consciously avoided privacy violations, which was supported by the prompt histories in the task sheets, showing no privacy-related content. This suggest that students demonstrated awareness of privacy considerations in their assignment work. To further elaborate on this, students were also asked whether they reflect on privacy beyond this assignment. Several students expressed that they generally relate to privacy issues, but that these are not necessarily considerations they reflect on extensively in their everyday use of digital services, such as AI tools and social media. One student stated that they would not want their face stored in an AI database, while another student added that during the introductory phase of new applications or social media platforms, users often do not reflect on the fact that uploaded images may be used as training data for AI systems. Students stated that using one's own face in AI-generated images, in principle, is an individual choice. However, several students pointed out that such choices are not always fully informed, partly due to insufficient or inaccessible information about terms of use. Some students also expressed that they were unaware of the existence of settings in various AI tools and social media platforms that allow users to control or limit how their personal data are used.

5.3.3 Students' reflections on misinformation in AI-generated content

A central topic of discussion throughout the period was misrepresentation and misinformation in using AI for advertising and other forms of content, particularly in cases where AI-generated material was not clearly disclosed. There was general agreement among the students that AI-generated advertising should be clearly labelled, for example in cases involving clothing and AI generated models, as they perceived that a lack of disclosure could contribute to uncertainty and confusion among audiences. Students argued that regulatory frameworks should require the labelling of AI-generated content, partly because consumers may develop unrealistic expectations, products may not correspond to the quality portrayed, and not all users are able to recognise that the content has been generated.

Another highlighted aspect concerns instances in which deepfakes are used to spread misleading information or fraudulent activities targeting vulnerable groups. One student noted having seen several videos in which celebrities were represented through deepfakes, and that it may be difficult to distinguish these from authentic videos. The student expressed particular concern about the potential consequences of using such technology towards children.

5.4 Redesign after the second intervention

5.4.1 Instructional design: Assignment execution and structure

After observing students work and what the students requested to be redesigned, these are what had to be revised:

- Clarity: A clearer walkthrough to be confident in the execution and some parts being too vague.
- Structured timeframes: more structured time frames to plan and outline their process.
- Extended time: More time for ideation and sketching to improve image generation outputs.

6 DISCUSSION

The aim of this study was to explore how the educational design, employing collaborative work with generative AI, enhanced students' development of AI and digital literacy, rather than measuring learning development across the interventions. The results indicate that students engaged actively with ethical and critical dimensions of AI use, particularly related to privacy, data use, and consent. Students' strong consensus around privacy suggests that the topics introduced in the design may function as an entry point for ethical reflection, aligning with core dimensions of AI-and-digital literacy frameworks (Long & Magerko, 2020; Norwegian Directorate for Education and Training, 2017).

Working with the assignment to design AI-generated advertisements facilitated discussions of image origins, consent, copyright, as well as strategies for responsible use. Through problem-solving and discussions, students demonstrated competencies in ethical AI use, source evaluation, and copyright awareness (Long & Magerko, 2020; Norwegian Directorate for Education and Training, 2017). Students' differing views on responsibility in AI generated advertising highlight that ethical judgement in this context is complex. These tensions can be understood as productive, as they indicate active engagement with ethical dilemmas rather than uncritical acceptance.

Students engaged extensively in peer discussion and collaborative reflection, often extending discussions beyond what was explicitly addressed in the educational design. The group discussions appeared to support understanding of AI-related issues, particularly when students were given space to articulate perspectives and build on one another's contributions.

Students' reflections were especially notable when discussing AI-related issues connected to relatable experiences from their digital everyday lives, such as misinformation, deepfakes, and unlabelled AI-generated content on social media and advertisements. This supports Dewey's understanding of learning as emerging through experience, inquiry, and problem-solving in response to real and relevant challenges (Dewey, 1916, 1938). When AI-generated outputs failed to meet students' expectations, peer discussion and iterative adjustments of prompt strategies became part of the learning process, indicating that reflection was embedded in the activity itself rather than occurring only after the task completion.

Overall, the results suggest that the educational design supports key aspects of AI and digital literacy by combining ethical reflections, peer interactions, and practice-based assignments with generative AI. The design aligns with dialogic co-creation approaches (Vartiainen et al., 2023), pragmatist views of learning through experiences and iteration (Dewey, 1916, 1938), active

participation (Heaton, et al. 2024) and the iterative logic of EDR (McKenney & Reeves, 2018). However, given the limited scope of the study, continued interventions are needed to further test and refine educational designs that support students' AI-and-digital literacy across educational practices to keep up with the ongoing technological development.

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